

3.6 Volatile Oils



What are volatile (essential) oils ?

- Volatile or essential oils, as their name implies, are volatile in steam.
- They differ entirely in both chemical and physical properties from fixed oils
- They are secreted in oil cells, in secretion ducts or cavities or in glandular hairs
- Usually associated with other substances such as gums and resins and themselves tend to resinify on exposure to air.
- significant quantities from raw plant materials from their organs: flowers, seeds, buds, leaves, fruits, wood, roots, bark, and twigs

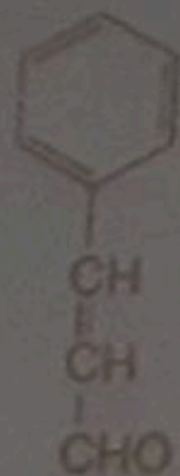
- Volatile oils are used for their :
 - flavoring (e.g. oil of lemon),
 - in perfumery (e.g. oil of rose)
 - For therapeutic purposes they are administered as inhalations (e.g. eucalyptus oil) orally (e.g. peppermint oil)
 - as gargles and mouthwashes (e.g. thymol)
 - Transdermally for aromatherapy (e.g. lavender, rosemary)

Composition of volatile oils

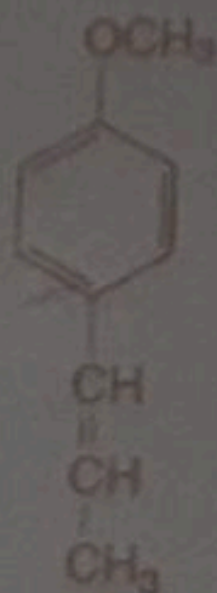
- Volatile oils are generally mixtures of **hydrocarbons** and their **oxygenated compounds** derivatives.
 - Hydrocarbons are the most abundant
- These oils may comprise a mixture of **more than 300 compounds**. Usually, their molecular weight is below **300 Da**.
 - Thus easily volatile
- The **odor and taste** of volatile oils is mainly determined by the alcohol soluble **oxygenated compounds**.
- Most of the compounds are of terpenoid origin but some are of aromatic derivatives
- Compounds can be cyclic or acyclic

- Different classes of compounds of essential oil constituents:

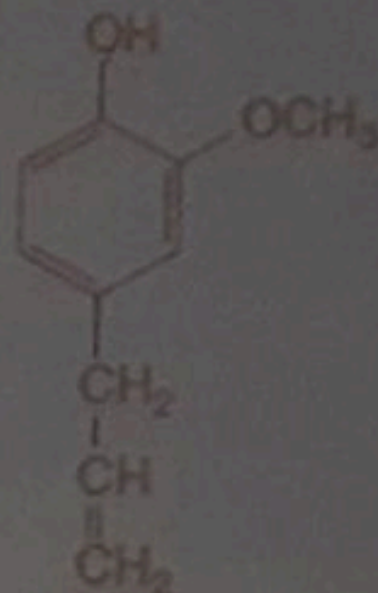
- terpenes and sesquiterpenes
- alcohols
- phenols
- ethers,
- carbonyl compounds such as aldehydes, ketones, amides, and esters. The chemical



Cinnamaldehyde



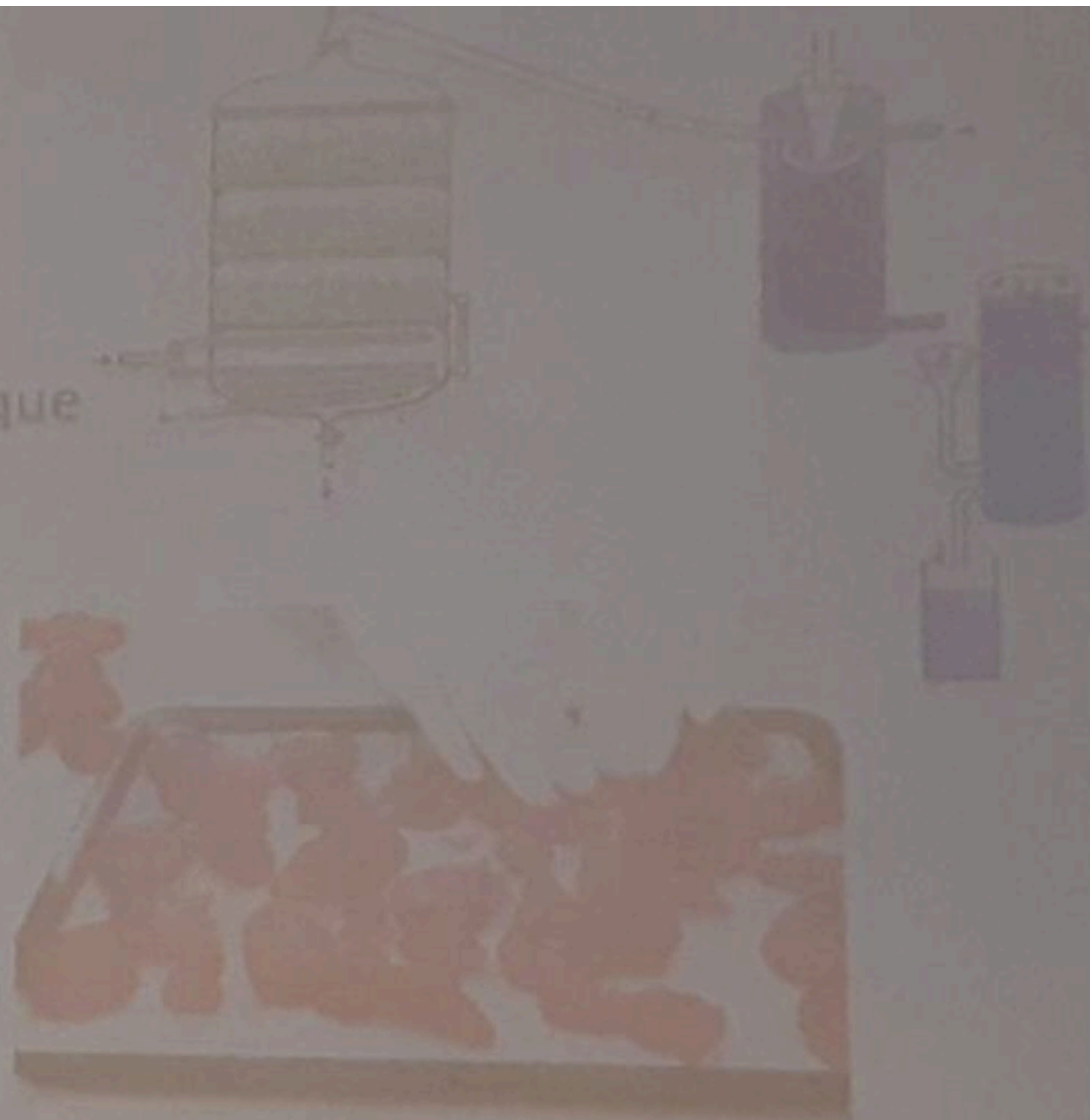
Anethole



Eugenol

Extraction

- Distillation is the widely used technique
 - Hydrodistillation or steam distillation
- Enfleurage
- Supercritical fluid extraction
 - Industrial scale



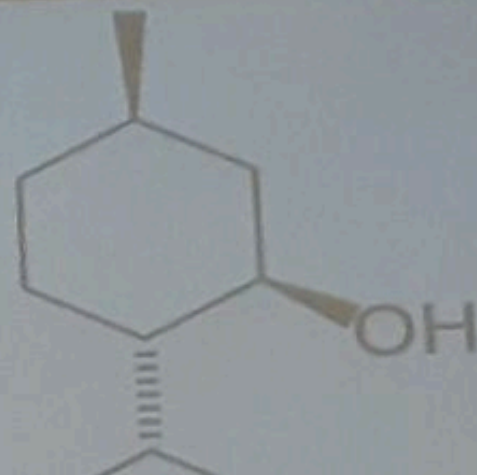
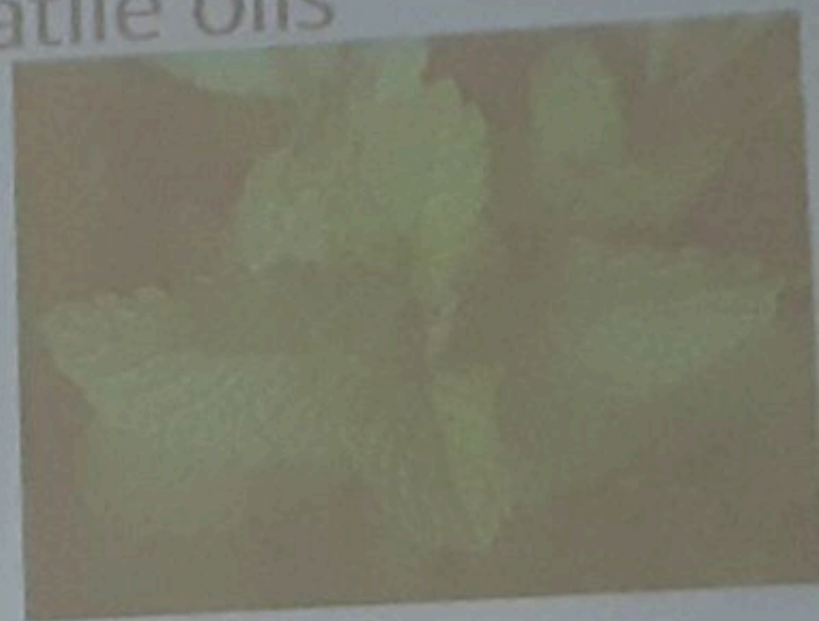
- Evaluation

- Preliminary : olfactory evaluation
- Physical: optical rotation, relative density and refractive index
- TLC , **GCMS (Golden standard)**

Some medicinally important volatile oils

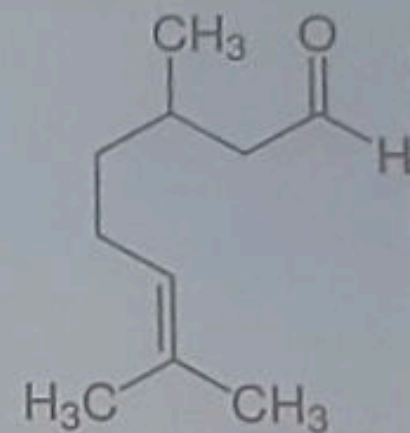
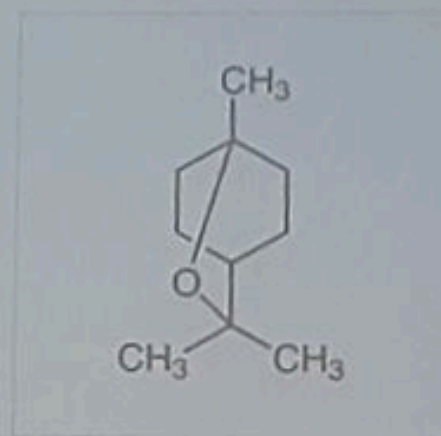
Peppermint Oil (*Mentha piperita*)

- Percentage yield: 1.2% of volatile oil.
- Steam distillation of flowering tops
- Two varieties
 - *M. piperita* var. *vulgaris* Sole ('black mint')
 - *M. piperita* var. *officinalis* Sole ('white mint').
- Major compound: **Menthol 30.0–55.0%**
- Topical counter irritant, carminative, flavoring



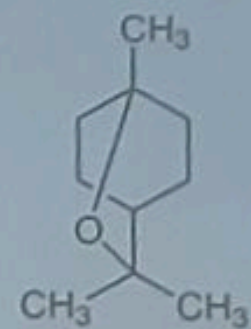
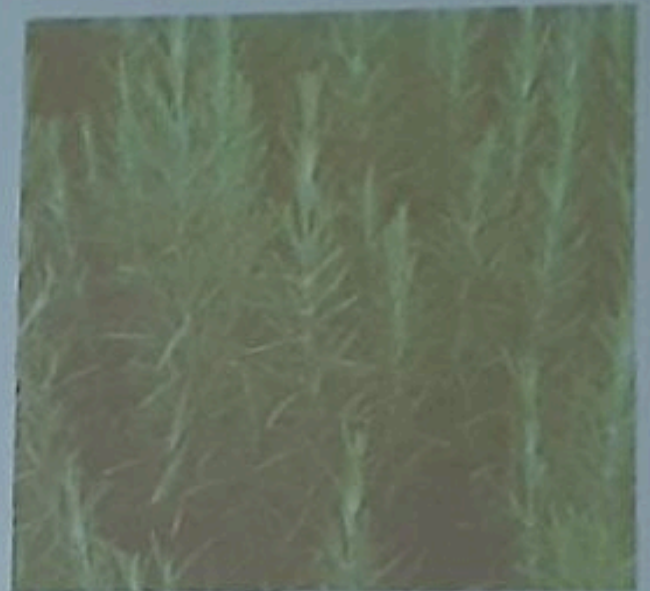
Eucalyptus Oil (*Eucalyptus globulus*)

- Oil of eucalyptus is a colourless or pale yellow liquid
- Percentage yield: 2%
- Major constituents: 70.0% of the oil is 1,8-cineole
 - *E. citriodora* contains an aldehyde **citronellal used in perfumery**
- Used to alleviating the symptoms of **nasopharyngeal infections**, for **treating coughs** and **as a decongestant**.



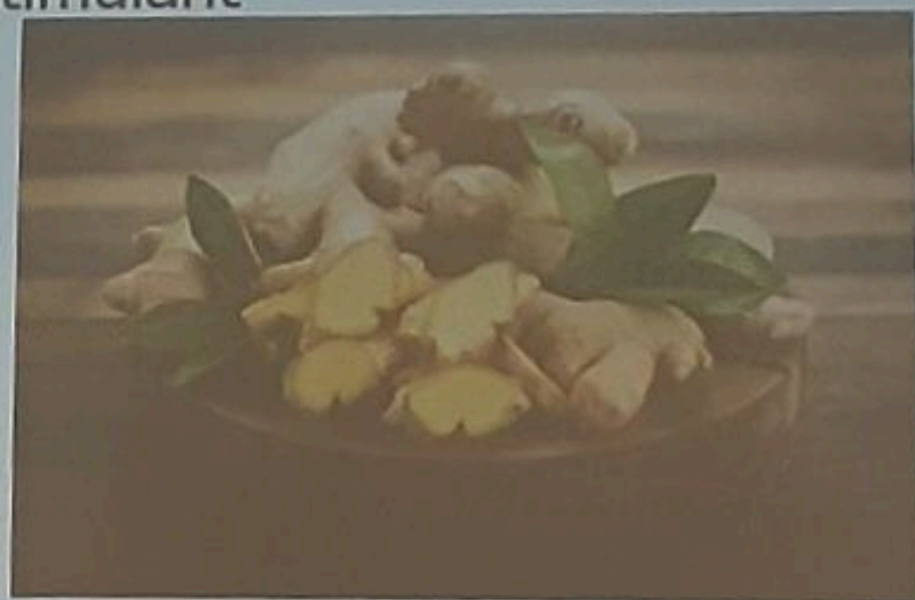
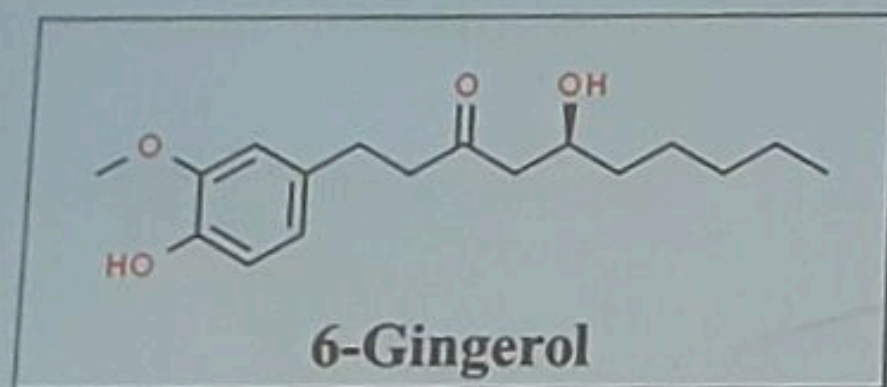
Rosemary Oil (*Rosmarinus officinalis* L.)

- Percentage yield: 1-2%
- colorless to pale yellow volatile oil
- Major constituents: **1,8-cineole, borneol**
- Used in: aromatherapy, perfumery industry
- It has **antibacterial and antispasmodic properties.**



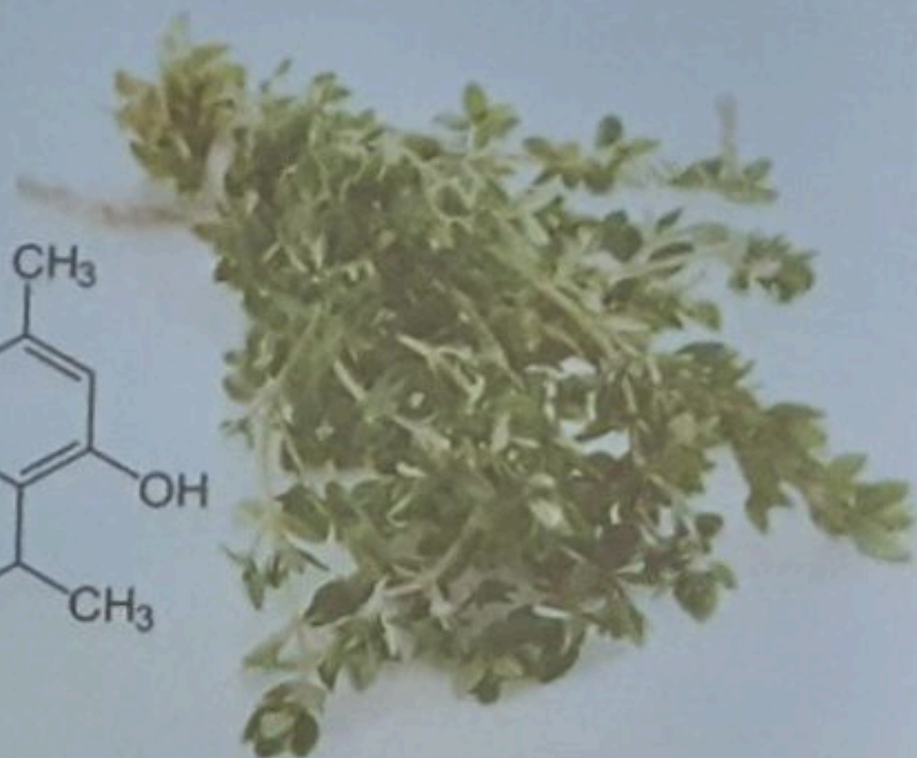
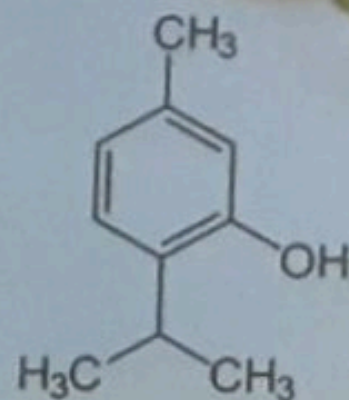
Ginger oil (*Zingiber officinale*)

- Different cultivates: Jamaica, China, India and Africa
- Percentage yeild: **1-2%**
- **6-Gingerol** is a major conistiuent
- **Potent antiemetic** for pregnant and patients with chemotherapy, antimicrobial, anti-inflammatory, GI stimulant



Thyme Oil (*Thymus vulgaris*)

- Yellow to dark reddish-brown
- Major constituents: **thymol (36–55%)**
- **treatment of:** respiratory infections, gastrointestinal problems and skin conditions requiring an antiseptic.



Reading assignments

- Resins
- Oleogum resins
- Aromatherapy